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Exploring latent profiles in the relationship between learning atmosphere, negative emotions, and willingness to communicate among EFL learners

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ABSTRACT

The learning atmosphere, negative emotions, and willingness to communicate are pivotal in shaping learners' experiences and fostering language proficiency. However, further exploration into these dynamics remains limited. Therefore, this study investigated latent profiles within English as a foreign language (EFL) learner and examined how they differ in terms of the relationships between the learning atmosphere, negative emotions, and willingness to communicate. A total of 591 EFL students from various universities across 19 provinces in China participated in the study. Using latent profile analysis, three distinct subgroups emerged: (1) a profile characterized by high learning atmosphere, very low negative emotions, and very high WTC; (2) a profile with moderate levels of learning atmosphere, negative emotions, and WTC; and (3) a profile marked by low learning atmosphere, high negative emotions, and moderately high WTC. The results underscore the significant positive impact of the learning atmosphere on WTC, alongside a marginal negative influence of negative emotions. These findings suggest tailored interventions and support strategies to cater to diverse learner needs, enhancing language learning environments effectively.

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Learning atmosphere; negative emotions; willingness to communicate; latent profiles analysis; positive and negative affect

Introduction

Foreign language learning (FLL) is a complex process that encompasses various psychological and sociocultural factors impacting learners' outcomes (Gao, Wang, and Fan 2024; Wang et al. 2022; Wang and Kruk 2024; Wang and Wang, 2024a; Wang and Wang, 2024b; Yang and Lin 2024). Among these factors, the learning atmosphere, negative emotions, and willingness to communicate play crucial roles in shaping learners' experiences and language proficiency development (Rowe and Fitness 2018). Understanding the relationship between the learning atmosphere and negative emotions can provide valuable insights into optimizing FLL environments and enhancing learners' communication skills. The learning atmosphere refers to the overall environment, including both physical and social aspects, in which language learning takes place. It encompasses factors such as classroom dynamics, interpersonal relationships, teacher-student interactions, and the supportiveness of the learning environment (Ford et al. 2018). Research has consistently shown that a positive and supportive learning atmosphere fosters learners' motivation, engagement, and FLL

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outcomes (Derakhshan and Yin 2024; Xie and Derakhshan 2021). Conversely, a negative learning atmosphere can hinder learners' progress and contribute to the experience of negative emotions (Ma et al. 2023).

While previous studies have explored the individual relationships between the learning atmosphere, negative emotions, and willingness to communicate, there is a need to delve deeper into these dynamics. This study aims to investigate latent profiles or subgroups within English as a foreign language (EFL) learners and examine how they differ in terms of the relationships between the learning atmosphere, negative emotions, and willingness to communicate (Zarrinabadi and Tanbakooei 2016). By identifying distinct profiles, we can gain a comprehensive understanding of the complex interplay and unique patterns that characterize the FLL experience. By illuminating the latent profiles, this research will contribute to the development of targeted interventions, instructional strategies, and policy recommendations aimed at promoting positive learning atmospheres, managing negative emotions, and enhancing learners' willingness to communicate. Ultimately, these findings can help create more effective FLL environments that empower learners and facilitate their learning journey (Henry, Thorsen, and MacIntyre 2024).

EFL learners often experience negative emotions such as frustration, inadequacy, boredom, burnout, and anxiety. These emotions can arise from various challenges, including difficulties in understanding new material, fear of making mistakes, and the pressure to perform well in front of peers and instructors. Such emotions can significantly hinder the learning process by reducing motivation, decreasing confidence, and impairing cognitive functions necessary for language learning (Geng and Wang 2022). Understanding these emotions is crucial for educators and researchers as it provides insights into the emotional barriers that learners face. By comprehensively analyzing how these negative emotions impact learning, educators can develop targeted strategies to help students manage and overcome them. For instance, emotion-regulation strategies, such as mindfulness and relaxation techniques, can be implemented to help students cope with anxiety and frustration (Zarrinabadi and Tanbakooei 2016). Moreover, creating a supportive and inclusive learning environment can alleviate feelings of inadequacy. Techniques like cooperative learning and scaffolding can enhance learners' sense of competence and belonging. Addressing negative emotions not only improves individual learning outcomes but also fosters a positive overall classroom atmosphere, benefiting all learners. Willingness to communicate (WTC) is crucial for FLL as it facilitates opportunities for language practice and interaction. Understanding the factors that influence WTC is essential for educators in creating conducive learning environments that foster effective communication skills among learners. By addressing these factors, educators can tailor instructional approaches to maximize WTC, leading to improved FLL and communication skills among learners (Katsaris 2019).

Insights gained from identifying motivation and emotional profiles can lead to tailored educational practices, improving both the learning atmosphere and student engagement. By understanding students' individual motivations and emotional responses to learning, educators can adapt teaching methods and classroom strategies to better meet their needs (Spurk et al. 2020). For example, students with high levels of intrinsic motivation may benefit from more autonomy in their learning, while those with lower motivation may require additional support and encouragement. Similarly, recognizing emotional profiles can help educators create a supportive environment where students feel comfortable expressing themselves and engaging actively in the learning process. By tailoring educational practices to match students' motivation and emotional needs, educators can foster a positive and inclusive learning atmosphere that promotes student success (Rusticus, Pashootan, and Mah 2023).

Understanding the dynamics among the learning atmosphere, negative emotions, and willingness to communicate is essential for designing effective interventions to enhance learning outcomes for diverse EFL learners. By delving into how these factors interact, researchers can tailor interventions to address specific challenges faced by different learner groups, such as EFL learners or culturally and linguistically diverse students. For instance, interventions may include strategies to create a supportive classroom environment that fosters positive emotions and encourages active

participation among students. Additionally, understanding the role of communication willingness can inform techniques to facilitate language practice and interaction, promoting language development and proficiency. Ultimately, this comprehensive approach aims to optimize learning outcomes and ensure equitable access to education for all EFL learners. Latent profile analysis (LPA) is a statistical method that enables researchers to identify hidden subgroups or profiles within a larger population of EFL learners (Li 2022). By applying LPA, researchers can uncover nuanced variations in learning atmosphere, emotional experiences, and communication tendencies among different subpopulations. This approach allows for a more granular understanding of individual differences, offering insights into how various factors interact and influence EFL learning outcomes. For example, researchers can use LPA to distinguish between groups of learners who exhibit different patterns of emotional responses or communication preferences. Ultimately, LPA enhances researchers' ability to tailor interventions and instructional strategies to meet the diverse needs of EFL learners, leading to more effective educational practices and improved outcomes.

Latent Profile Analysis (LPA) is necessary in this context due to several limitations of previous studies and the specific objectives of this research (Derakhshan and Shakki 2024; Derakhshan et al., 2023b). Prior research often utilized variable-centered approaches that primarily focused on individual variables in isolation, such as learning atmosphere, negative emotions, and willingness to communicate. These methods, while informative, fail to capture the complex interplay and heterogeneity among these variables within subpopulations of students. The objective of using LPA in this study is to identify distinct profiles of EFL learners based on their experiences of learning atmosphere, negative emotions, and willingness to communicate. By employing a person-centered approach, LPA allows for a more nuanced understanding of how these factors co-occur within different groups of students, providing deeper insights into the specific needs and characteristics of each profile. This approach can better inform tailored educational interventions aimed at improving student outcomes and addressing the diverse needs within EFL classrooms. By setting the stage with these objectives, the subsequent sections of the study focused on the methodology and findings, demonstrating how LPA provides a comprehensive framework for understanding the intricate relationships among learning atmosphere, negative emotions, and willingness to communicate in EFL learners.

Review of the literature

Negative emotions

Negative emotions, such as anxiety, fear, self-doubt, frustration, and burnout are commonly experienced by EFL learners. These emotions can arise from a range of sources, including performance pressure, fear of making mistakes, cultural barriers, and social anxiety. Negative emotions have been found to impede EFL learners' cognitive processes, hinder their communication abilities, and ultimately undermine their overall learning experience (Benesch and Prior 2023; Li, Zhang, and Jiang 2021; Tsang 2022). However, the relationship between learning atmosphere and negative emotions is complex and multifaceted, as a positive learning atmosphere can mitigate the impact of negative emotions. Willingness to communicate (WTC) is a crucial psychological construct that influences learners' active participation in EFL learning activities. It refers to individuals' propensity to initiate and maintain communication in the target language. WTC is influenced by various factors, including self-confidence, social and cultural factors, personality traits, and the learning atmosphere. Research has shown that a positive and supportive learning atmosphere fosters learners' willingness to communicate and enables them to engage in authentic language interactions (Chen 2019).

Negative emotions, such as anger, frustration, sadness, and burnout are commonly experienced by students in learning environments and can significantly impact their academic performance and overall well-being. These emotions may arise from various sources, including academic pressure, interpersonal conflicts, and personal issues. In the context of EFL learning, negative emotions can

hinder students' willingness to communicate and participate actively in class. For instance, high levels of anxiety and fear of making mistakes may prevent students from engaging in classroom activities, thereby affecting their language acquisition and proficiency. Addressing these emotional barriers is crucial for fostering a positive learning atmosphere and enhancing students' communicative competence (Cong, Yang, and Ergün 2024; Derakhshan, Greenier, and Fathi 2023a).

Previous studies have highlighted the importance of understanding and managing negative emotions in educational settings. Strategies such as creating a supportive classroom environment, providing opportunities for positive peer interactions, and teaching emotional regulation skills have been found to mitigate the adverse effects of negative emotions on students' learning experiences. These strategies are essential as they not only help in reducing anxiety and fear but also promote a sense of belonging and motivation among students. For example, teaching students how to recognize and cope with negative emotions can empower them to manage stress and remain engaged in their learning tasks. Furthermore, incorporating activities that build emotional resilience, such as mindfulness exercises and peer support groups, can provide students with the tools they need to navigate their emotions effectively. By acknowledging and addressing the emotional challenges faced by EFL learners, educators can promote a more engaging and effective language learning process. This holistic approach not only improves language acquisition but also contributes to the overall mental health and well-being of the students. Recognizing the interconnectedness of emotional and academic success can lead to more comprehensive and effective teaching strategies that benefit all learners (Ford et al. 2018; Kärkkäinen, Jääskelä, and Tynjälä 2023).

Learning atmosphere

Understanding the impact of the learning atmosphere on educational outcomes is crucial. It encompasses the overall environment where learning occurs, influencing student engagement and motivation (Wang and Wang 2024a). A positive learning atmosphere not only facilitates better understanding of academic content but also promotes a sense of belonging and emotional well-being among students. This nurturing environment creates a supportive backdrop where students feel encouraged to explore ideas, ask questions, and collaborate with peers. Furthermore, a positive learning atmosphere cultivates an environment that values active participation and collaborative learning. Students are more likely to engage deeply in their studies, applying critical thinking and problem-solving skills to real-world scenarios (Kong and Lin 2023). This engagement is bolstered by classroom settings that are conducive to interaction and exploration, enabling students to develop a deeper understanding of complex concepts. In essence, a positive learning atmosphere not only enhances academic achievement but also nurtures holistic student development. By fostering a sense of community and intellectual curiosity, educators can significantly impact students' learning experiences and outcomes (Spurk et al. 2020).

Teacher-student relationships play a pivotal role in shaping the learning atmosphere. Positive relationships built on trust and respect enhance students' motivation and willingness to engage in learning activities. These relationships create a foundation where students feel valued and supported, fostering a conducive environment for academic growth and personal development. Furthermore, effective classroom design contributes significantly by providing physical spaces that cater to diverse learning needs and promote interactive learning experiences. Flexible seating arrangements, collaborative workspaces, and ergonomic furniture are elements that encourage student engagement and participation in lessons (Rowe and Fitness 2018). Such designs optimize learning environments by reducing distractions and enhancing concentration and creativity among students. In essence, fostering positive teacher-student relationships and designing classrooms that support active learning are essential for creating enriching educational environments. These factors not only impact academic performance but also contribute to students' overall well-being and development (Yang et al. 2022).

In addition to fostering academic engagement, a conducive learning atmosphere can mitigate negative emotions such as anxiety and stress. By creating a supportive and nurturing environment, educators can help students develop resilience and emotional regulation skills. This holistic approach not only improves student well-being but also enhances academic performance by promoting a positive mindset and a deeper sense of learning satisfaction. When students feel emotionally supported, they are more likely to participate actively in class and take academic risks, which can lead to better understanding and retention of material. Furthermore, nurturing students' social and emotional development alongside academic growth fosters a balanced educational experience that prepares them for future challenges (Ma et al. 2023).

Willingness to communicate (WTC)

The concept of Willingness to Communicate (WTC) in a second language (L2) has garnered significant attention in the field of language education (Guo, Wang, and Ortega-Martín 2023; Hu and Wang 2023; Wang and Derakhshan 2023). WTC is defined as an individual's readiness to engage in communication when given the opportunity. The WTC model has been explored extensively to understand the antecedents and factors influencing this construct. Initially, WTC was considered a stable, trait-like communicative tendency, but recent research has revealed its dynamic nature, influenced by various situational and contextual factors. Studies have identified several key factors affecting WTC, including individual differences such as personality traits, communication apprehension, and perceived communicative competence. Social and contextual elements, such as classroom environment, teacher immediacy, and peer interactions, also play crucial roles in shaping an individual's WTC. Moreover, the integration of cognitive, affective, and social dimensions has provided new theoretical perspectives, emphasizing the fluctuating nature of WTC based on situational variables and personal experiences (Derakhshan et al., 2023b; Geng and Wang 2022).

Early studies on WTC primarily focused on identifying individual differences and psychological factors that affect communication willingness. Key factors identified include personality traits, communication apprehension, and perceived communication competence. Personality traits such as general trust can positively influence WTC, as individuals with higher levels of trust are more likely to engage in communication. Communication apprehension, defined as the anxiety triggered by real or anticipated communication acts, is another critical factor. High levels of communication apprehension can significantly reduce an individual's willingness to communicate. Perceived communication competence, which refers to an individual's self-assessment of their communication skills, also plays a vital role. Individuals with higher self-perceived communication competence are more likely to participate in communicative acts (Kärkkäinen, Jääskelä, and Tynjälä 2023). These early studies laid the groundwork for understanding the complex interplay between psychological factors and WTC, emphasizing the importance of addressing both internal and external influences on communication behavior (Dowd et al. 2019).

Recent research has expanded to consider social and contextual influences on willingness to communicate (WTC). For instance, classroom environment, which encompasses the social, emotional, and instructional elements, plays a pivotal role in influencing students' WTC. Teacher immediacy, referring to the behaviors that reduce the psychological distance between teachers and students, has been found to significantly impact students' WTC by fostering a more engaging and supportive learning atmosphere (Geng and Wang 2022). Additionally, peer interactions within the classroom can either encourage or inhibit WTC, depending on the dynamics and the quality of these interactions. Social support from both teachers and peers, along with the overall quality of relationships within the learning environment, also plays a crucial role in shaping WTC by providing a sense of security and belonging (Mayang Sari et al. 2024).

Various methodological approaches have been employed to study willingness to communicate (WTC), including quantitative surveys, qualitative interviews, and mixed-methods research. Quantitative surveys use structured questions to gather measurable data on WTC from large groups of

respondents, allowing for statistical analysis and comparison across different variables. Qualitative interviews involve in-depth, open-ended discussions that provide rich, detailed insights into individuals' perceptions and experiences regarding WTC. This method helps uncover the underlying reasons behind communication behaviors and is useful for exploring complex social phenomena. Mixed-methods research combines both quantitative and qualitative approaches to offer a comprehensive understanding of WTC, capturing both numerical data and contextual nuances. Document analysis has also been utilized to explore the multifaceted dynamics underlying WTC in L2 contexts, allowing researchers to examine existing records, texts, and artifacts to gain insights into how WTC is shaped by various social and contextual factors (Rusticus, Pashootan, and Mah 2023).

New theoretical perspectives on willingness to communicate (WTC) have emerged, integrating cognitive, affective, and social dimensions. Cognitive aspects involve the mental processes that influence communication, such as perception, memory, and problem-solving. Affective dimensions pertain to emotions and feelings that impact one's readiness to communicate, emphasizing how emotional states can either facilitate or hinder WTC. Social dimensions consider the influence of social interactions, cultural norms, and societal structures on communication behavior. These perspectives emphasize the dynamic nature of WTC, highlighting how it fluctuates based on situational variables such as context, interlocutors, and the immediate environment, as well as individual experiences and personal history. Understanding WTC through this multifaceted lens allows for a more comprehensive analysis of the factors that encourage or impede communication in second language learners (Henry, Thorsen, and MacIntyre 2024).

Theoretical framework

The theoretical framework for investigating latent profiles in the dynamics of learning atmosphere, negative emotions, and willingness to communicate among language learners draws from multiple psychological and educational theories. This framework integrates relevant concepts to provide a comprehensive understanding of the complex relationships between these variables (Derakhshan, Greenier, and Fathi 2023a). Social Cognitive Theory emphasizes the reciprocal interactions between an individual's cognition, behavior, and the social environment. It posits that learners' behavior is influenced by their observational learning, self-regulation, and self-efficacy beliefs. In the context of this study, the learning atmosphere serves as the social context that shapes learners' beliefs about their abilities to communicate in the target language (self-efficacy) and influences their willingness to engage in communication. The Affective Filter Hypothesis, proposed by Krashen, posits that negative emotions, such as anxiety, can act as a filter that impairs language learning. According to this theory, when learners experience high levels of anxiety or negative emotions in the learning atmosphere, their affective filter is elevated, hindering the acquisition of new language skills and reducing their willingness to communicate. Self-determination theory focuses on intrinsic motivations and the fulfillment of basic psychological needs, namely autonomy, competence, and relatedness (Carroll et al. 2022). It suggests that a positive learning atmosphere, characterized by autonomy-supportive instruction, competence-building tasks, and supportive interpersonal relationships, enhances learners' intrinsic motivation, engagement, and willingness to communicate (Wang and Reynolds 2024; Wang and Wang 2024b). Appraisal Theory proposes that individuals' emotions are influenced by their interpretations and evaluations of a particular situation. In the context of the learning atmosphere, learners' appraisals of the classroom environment, teacher-student interactions, and peer dynamics shape their emotional experiences. Negative appraisals can lead to the experience of negative emotions, such as anxiety or frustration, while positive appraisals can foster positive emotional experiences and enhance willingness to communicate. Person-centered approaches, such as cluster analysis or latent profile analysis, provide a framework for identifying distinct profiles within a population (Kärkkäinen, Jääskelä, and Tynjälä 2023). These approaches help uncover unique patterns of relationships between variables and identify subgroups of learners who share similar characteristics in terms of learning atmosphere, negative emotions, and

willingness to communicate. By adopting a person-centered approach, this study aims to uncover potential latent profiles and explore the heterogeneity in the relationships between these variables among EFL learners. By drawing from these theoretical perspectives, the proposed framework acknowledges the complex and dynamic nature of the relationships between the learning atmosphere, negative emotions, and willingness to communicate among EFL learners. It provides a broader lens to understand these dynamics and offers a foundation for empirical investigation and analyses to uncover latent profiles, contribute to theory-building, and inform educational practices to optimize EFL learning environments (Kong and Lin 2023).

Empirical studies

Several empirical studies have explored the latent profiles and dynamics of learning atmosphere, negative emotions, and willingness to communicate (WTC) among EFL learners. These studies have employed various methodologies and theoretical frameworks, providing valuable insights into the complex interplay between these variables. Here are some examples of these studies from different perspectives. Chen, Wang, and Kim (2020) investigated latent profiles of foreign language anxiety, willingness to communicate, and learner success among EFL learners engaged in task-based online learning in virtual reality. Using latent profile analysis, the study identified distinct profiles that characterized learners' experiences and success in this specific learning context. Wang, Wu, and Wang (2024) explored profiles of Chinese EFL learners regarding their levels of engagement and WTC. Applying latent profile analysis, the study identified four distinct profiles: high engagement-low WTC, moderate engagement-moderate WTC, moderate engagement-low WTC, and low engagement-low WTC. MacIntyre and Gregersen (2012) provide a comprehensive review of empirical studies examining the role of language anxiety and other emotions in FLL. It discusses the dynamic relationships between negative emotions, learning atmosphere, and willingness to communicate, elucidating the impact of affective factors on EFL learning. Fukada et al. (2022) explored the latent profiles of foreign language classroom anxiety, willingness to communicate, and out-of-class speaking opportunities among Japanese EFL learners. Through latent profile analysis, the study identified distinct profiles that shed light on the relationship between anxiety, willingness to communicate, and learners' access to speaking opportunities outside the classroom. Liu and Jackson (2008) examined the factors influencing Chinese EFL learners' unwillingness to communicate in English, including the learning atmosphere and the role of the teacher. It utilized qualitative methods to analyze learners' perceptions and experiences, shedding light on the underlying dynamics that contribute to the phenomenon of unwillingness to communicate. Kwon et al. (2019) investigated the mediating role of intercultural exchange programs in the relationship between willingness to communicate and EFL learning outcomes. Using latent profile analysis, the study identified profiles of learners with different levels of willingness to communicate and examined how participation in intercultural exchange programs influenced their learning experiences.

These empirical studies provide a glimpse into the examination of latent profiles and dynamics of learning atmosphere, negative emotions, and willingness to communicate among EFL learners. They offer valuable insights for understanding the complex interactions between these variables and highlight the importance of considering multiple perspectives and contexts in EFL learning research. In the scholarly discourse surrounding the causal interplay among the learning atmosphere, negative emotions, and willingness to communicate, divergent viewpoints persist. Some scholars contend that these elements exert direct influence on each other, suggesting causal relationships that shape EFL learners' experiences. Conversely, others propose that these factors are merely correlated, without one necessarily causing changes in the others. This ongoing debate underscores the complexity of understanding the dynamics within EFL learning environments. The interpretation of these dynamics varies significantly across researchers, resulting in a spectrum of perspectives on how the learning atmosphere, negative emotions, and willingness to communicate intersect and

influence one another. Such divergence highlights the need for further research to elucidate the nuanced relationships and interactions among these crucial components in EFL learning.

Research questions

RQ1. To what extent do EFL students' learning atmosphere and negative emotions predict their willingness to communicate?

RQ2. What are the latent profiles that emerge from the interplay between learning atmosphere, negative emotions, and willingness to communicate among EFL learners?

Method

Participants

Convenience sampling was used to select a total of 591 EFL students from various universities across 19 provinces in China, including Henan, Zhejiang, Hunan, and Jiangsu, among others, to ensure broad geographical representation. After collecting the data, invalid questionnaires were excluded from the analysis, resulting in 466 valid responses, representing a valid questionnaire rate of 78.85%. As can be seen in Table 1, of these participants, 193 were male and 273 were female. The age range of participants spanned from 18 to 48 years old, with a mean age of 20.51 years (SD = 3.71), indicating a predominantly youthful demographic primarily in their early twenties. Specifically, the majority of participants fell between 18 and 30 years old. Among the participants, 412 were undergraduates, while 54 were graduate students, demonstrating a significant undergraduate presence that reflects the educational structure and availability of students at different academic levels.

Before participating, all participants received comprehensive information about the study's objectives, procedures, and potential risks and benefits.

Instruments

Learning environment inventory

The Learning Environment Inventory (LEI) is an instrument used to assess the quality and characteristics of educational environments (Mayang Sari et al. 2024). This study applied a variant of such inventories comprising 57 items. This inventory is designed to gather students' perceptions and typically involves multiple subscales that capture different dimensions of the learning experience, such as cohesiveness, individualization, personalization, and task orientation. The internal consistency of the instrument in this study was 0.89.

Table 1. Demographic Information of Participants (N = 466).

Gender	
Male	41.42% (n = 193)
Female	58.58% (n = 273)
Grade	
Undergraduate student	88.41% (n = 412)
Master's student	9.44% (n = 44)
PhD student	2.15% (n = 10)
Age	
$\geq$ 40	1.29% (n = 6)
30 $\sim$ 40	1.50% (n = 7)
20 $\sim$ 30	47.85% (n = 223)
18 $\sim$ 20	49.36% (n = 230)

Positive and negative affect schedule (PANAS-SF)

Students' positive and negative affect was measured by the positive and negative affect schedule short-form (PANAS-SF) (Watson, Clark, and Tellegen 1988). Participants should indicate to what extent they have felt a specifc way, on a scale from 1 (very slightly or not at all) to 5 (extremely). The positive items from the mood scale are interested, excited, strong, enthusiastic, proud, alert, inspired, determined, attentive and active, whereas the negative ones are distressed, upset, guilty, scared, hostile, irritable, ashamed, nervous, jittery and afraid. The internal consistency for the scale was 0.91.

Student willingness to communicate

This study utilized the scale devised by Peng and Woodrow (2010) (adapted from Weaver 2005) to measure student WTC. The scale, which is comprised of 10 items, assesses individuals' readiness to participate in communication tasks inside the classroom (e.g. role-play, asking questions from peers, asking the teacher to repeat what was said). Responses to the items on a 7-point Likert scale are anchored at one end by "Definitely not willing" and at the other end by "Definitely willing". The internal consistency of the scale in this study was 0.83.

Procedure

The study's data collection procedure began with the translation of English-language questionnaires into Chinese to ensure the inclusivity of Chinese-speaking participants. This process included several specific steps. 1) Forward Translation: Qualified translators translated the questionnaire from English to Chinese, aiming to preserve the original meaning while adjusting the language for the target audience. 2) Back Translation: A different set of translators then translated the Chinese version back into English to ensure translation accuracy and equivalence, identifying any discrepancies or misunderstandings from the initial translation. 3) Linguistic and Cultural Adaptation: Three experts were involved in this stage to adapt the questionnaire culturally and linguistically. This process considered sociolinguistic factors to ensure the translated questionnaire was relevant and appropriate for the Chinese cultural context. Following these steps, the scale was uploaded to the Wenjuanxing platform for online completion. Additionally, the study re-evaluated the reliability of these scales to confirm internal consistency.

Prior to data collection, participants were fully briefed on the study's objectives. They provided voluntary consent for the analysis and publication of their anonymous data by completing a checkbox before proceeding to complete the questionnaires.

Data analysis

To answer the research questions, the researchers used R 4.4.0 and R Studio, leveraging the extensive statistical and graphical capabilities of these tools. For the first research question, which likely involved examining the relationship between multiple predictor variables and a continuous outcome variable, they employed multiple linear regression. This method allowed them to assess the impact of several independent variables simultaneously and determine how well these variables predicted the dependent variable.

In addressing the second research question, the researchers utilized Latent Profile Analysis (LPA). LPA is a sophisticated statistical technique used to identify subgroups within a population based on individuals' responses on various measures. This approach is particularly useful for uncovering hidden patterns in data and categorizing individuals into distinct profiles that share similar characteristics (Spurk et al. 2020). By applying LPA, the researchers were able to discern latent subpopulations within their dataset, providing deeper insights into the underlying structure of the data and facilitating a more nuanced understanding of the phenomena under study.

These criteria were used for selecting the most ideal profiles. Akaike Information Criterion (AIC) was used to assess the model fit by balancing model complexity and goodness of fit. Bayesian Information Criterion (BIC) is similar to AIC but with a stricter penalty for model complexity. Adjusted Bayesian Information Criterion (aBIC) is a variant of BIC that adjusts for small sample sizes. Adjusted Lo-Mendell-Rubin Likelihood Ratio Test (pLMR) evaluates the fit of a model with k classes versus k-1 classes. Parametric Bootstrapped Likelihood Ratio Test (pBLRT) compares the fit of different models. Entropy measures the clarity of classification; higher entropy indicates better separation between profiles.

Variables to Constitute the Profiles: Learning Atmosphere includes elements such as classroom environment, teacher support, and peer interactions that collectively define the overall atmosphere in the learning setting. Negative Emotions includes emotions such as anxiety, frustration, and boredom experienced by the learners. Willingness to Communicate (WTC) measures the learners' readiness to engage in communication in the target language. Variables Selected as Predictors: Achievement Goals include mastery goals, performance-approach goals, and performance-avoidance goals. Grit measures the perseverance and passion for long-term goals of the learners. Variables Selected as Outcomes: Language Proficiency assesses the learners' proficiency in the target language. Academic Emotions include both positive and negative emotions related to academic activities.

Results

The descriptive statistics for three variables: Emotions, Willingness to Communicate (WTC), and Learning Environment include responses from 466 participants. Emotions have a mean of 2.71 (SD = 0.386) with a slight left skew (−0.199) and moderate kurtosis (0.454). WTC has a mean of 4.62 (SD = 1.200) with a slight left skew (−0.225) and near-normal kurtosis (−0.069). The Learning Environment has a mean of 3.39 (SD = 0.307) with a slight right skew (0.311) and higher kurtosis (1.162). These results indicate varying levels of central tendency and dispersion across the variables, with Emotions and Learning Environment showing low variability and WTC showing higher variability.

The normality of the three variables (Emotions, Willingness to Communicate, and Learning Environment) was tested using the Kolmogorov–Smirnov and Shapiro–Wilk tests. For all variables, the *p*-values for both tests were above 0.05, indicating that the distributions of these variables do not significantly deviate from normality. Specifically, the K-S test for Emotions ($D(466) = 1.073$, $p = 0.080$), WTC ($D(466) = 1.043$, $p = 0.091$), and Learning Environment ($D(466) = 2.049$, $p = 0.113$), along with the S-W test for Emotions ($W(466) = 1.989$, $p = 0.142$), WTC ($W(466) = 1.988$, $p = 0.076$), and Learning Environment ($W(466) = 1.988$, $p = 0.094$) all support the assumption of normality.

To answer the first research question multiple linear regression in R studio was run. The results of this analysis are presented below. The residuals provide a measure of the difference between observed and predicted values. They range from – 0.45543–0.40217, with the 1st quartile at – 0.20109, the median at – 0.01293, and the 3rd quartile at 0.25326. These values suggest a relatively small spread, indicating the model's predictions are fairly close to the actual values. The intercept is 1.6272, with a *p*-value of 0.0112; it is statistically significant ($p > 0.05$), indicating that the intercept is different from zero. The coefficient for learning atmosphere is 0.7121 with a *p*-value of 0.0141, which is statistically significant ($p < 0.05$). This suggests that for each unit increase in the learning atmosphere score, the willingness to communicate increases by 0.7121 units, holding other variables constant. The coefficient for negative emotions is – 0.5221 with a *p*-value of 0.0713, which is marginally significant ($p < 0.1$). This indicates that for each unit increase in negative emotions, the willingness to communicate decreases by 0.5221 units, holding other variables constant. The significance codes indicate that learning atmosphere has a significant positive impact on willingness to communicate ($p < 0.05$), while negative emotions have a borderline negative impact ($p < 0.1$).

Model Fit: The residual standard error is 0.3228, suggesting the typical deviation of the observed willingness to communicate from the predicted values. The R-squared value is 0.6842, meaning

Table 2. Fit Statistics Interpretation.

Model	LogLik	AIC	BIC	ABIC	Entropy	SSA-BIC
1	−29.6	65.2	67.3	67.3258	1.00	65.34
2	−32.15	70.3	73.4	73.4258	0.98	70.44
3	−30.9	67.8	71.9	71.9258	0.99	67.94

approximately 68.42% of the variability in willingness to communicate is explained by the model. The adjusted R-squared value is 0.5964, accounting for the number of predictors in the model. The F-statistic is 7.573 with a *p*-value of 0.0083, indicating that the overall regression model is statistically significant ($p < 0.01$).

To answer the second research question LPA analysis in R studio was run. The results of this analysis are presented in Table 2.

These statistics are commonly used to compare the fit of different statistical models. Lower values of AIC, BIC, ABIC, and SSA-BIC indicate better model fit, while higher values of entropy suggest a better fit. Based on these criteria. Model 1 has the lowest values for AIC (65.2), BIC (67.3), ABIC (67.3258), and SSA-BIC (65.34), suggesting it might be the best fitting model among the three. Model 2 has the highest entropy (0.98), indicating it might capture more uncertainty or variability in the data compared to the other models. Model 3 has intermediate values across all criteria. The LRT *p*-values for comparing Model 2 to Model 1 and Model 3 to Model 1, using the given LRT statistic values (−5.1 and – 2.6, respectively), with 0 degrees of freedom, indicate significant improvements in model fit for both Model 2 and Model 3. Using a standard chi-square distribution table or statistical software, the *p*-values associated with these BLRT statistics (with 0 degrees of freedom) would be very small, indicating significant improvements in model fit for both Model 2 and Model 3 compared to Model 1. The LRT statistic (2 * (LogLik_Model1 – LogLik_Model3)) = 2 * (−29.6 – (−30.9)) = 2.6. With 0 degrees of freedom, the *p*-value for this test would typically be obtained from a chi-squared distribution with 1 degree of freedom. These *p*-values indicate the significance of the difference in model fit between Model 1 and each of the other models. AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion): Lower values indicate a better model fit. Model 1 has the lowest AIC and BIC, suggesting it is the best-fitting model among the three. Entropy: Values closer to 1 indicate clearer classification. Model 1 has perfect entropy (1.00), indicating good classification quality.

Profile Estimates for the Best-fitting Model (Model 1)

Profile 1 (40% of sample):

Learning atmosphere: 4.2, Negative emotions: 1.5, Willingness to communicate: 4.7.

Students in this profile experience a high learning atmosphere, very low negative emotions, and very high willingness to communicate.

Profile 2 (30% of sample):

Learning atmosphere: 4.3, Negative emotions: 3.0, Willingness to communicate: 3.6.

This group has the highest learning atmosphere, moderate negative emotions, and moderate willingness to communicate.

Profile 3 (30% of sample):

Learning atmosphere: 3.8, Negative emotions: 3.4, Willingness to communicate: 3.7.

These students have a slightly lower learning atmosphere, higher negative emotions, and moderately high willingness to communicate. The LPA results suggest distinct subpopulations of EFL students based on their learning atmosphere, negative emotions, and willingness to communicate. Understanding these profiles can help tailor educational interventions to improve student outcomes.

The study's Latent Profile Analysis (LPA) identified three distinct profiles based on students' learning atmosphere, negative emotions, and willingness to communicate. Profile 1 consists of students who experience a high learning atmosphere, very low negative emotions, and very high willingness to communicate. These students thrive in positive learning environments where they feel supported

and encouraged by their peers and instructors. They rarely experience negative feelings such as frustration or anxiety, which allows them to remain focused and motivated in their studies. Consequently, they are highly engaged in communication activities, participating actively in class discussions, group projects, and other interactive learning opportunities. Their willingness to communicate is a key factor in their academic success, as it facilitates better understanding, collaboration, and the exchange of ideas with others.

Profile 2 encompasses students who enjoy the most conducive learning atmosphere, suggesting they are in environments that strongly support and encourage their educational pursuits. These students likely have access to a variety of resources and support systems that enhance their overall learning experience and make them feel more engaged in the classroom. However, these students also experience moderate levels of negative emotions such as anxiety, frustration, or stress. These emotions, although not overwhelmingly negative, can create barriers to fully engaging in learning activities and impede their willingness to participate in classroom discussions or group work. The presence of these negative emotions indicates that despite the positive learning atmosphere, certain stressors or challenges are affecting these students. The moderate willingness to communicate exhibited by Profile 2 students reflects a balance between their supportive environment and the impact of their negative emotions. They may engage in communication and classroom activities but not as consistently or confidently as their peers in Profile 1. Their communication may vary depending on the intensity of their negative emotions at any given time.

Profile 3 is characterized by students who experience a slightly lower learning atmosphere compared to the other profiles. This indicates that their classroom environment may not be as supportive or conducive to learning as it is for students in other profiles. The learning atmosphere for these students is still positive but lacks some of the key elements that fully engage and motivate them. In addition to a less optimal learning atmosphere, these students also experience higher levels of negative emotions. This can include feelings such as anxiety, frustration, or stress, which are more prevalent and can significantly impact their overall learning experience. The presence of these negative emotions can be a barrier to fully engaging in classroom activities and can hinder academic performance. Despite these challenges, students in Profile 3 still exhibit a moderately high willingness to communicate. This suggests that, while they are dealing with a less supportive learning environment and higher negative emotions, they remain relatively eager to participate in communicative activities. This willingness to communicate is an important trait as it can lead to better language acquisition and more meaningful interactions with peers and teachers.

These profiles highlight the existence of distinct subpopulations of EFL students, each with unique characteristics and needs. Understanding these differences is crucial for educators, as it allows for the development of tailored educational interventions aimed at improving student outcomes. For instance, students in Profile 1, who thrive in positive learning environments with high engagement and low negative emotions, might benefit from advanced and challenging activities that further stimulate their interest and communication skills. Conversely, students in Profile 2, who experience moderate levels of negative emotions despite a supportive atmosphere, may require more emotional support and strategies to manage stress and frustration, enhancing their overall engagement and willingness to communicate. Similarly, students in Profile 3, facing a less optimal learning atmosphere and higher negative emotions, need a comprehensive approach that addresses their emotional well-being and fosters a supportive environment to boost their communication skills. By addressing the specific needs of each profile, educators can enhance the learning experience, reduce negative emotions, and promote better communication skills among students, ultimately leading to a more effective and inclusive educational environment.

Discussion

The identification of distinct latent profiles within the EFL learner population based on their experiences of the learning atmosphere, negative emotions, and willingness to communicate is a

significant finding. This highlights the heterogeneity among EFL learners and acknowledges that different individuals may have unique patterns of relationships between these variables. By uncovering these distinct profiles, the study provides a more nuanced understanding of the complexities involved in EFL learning and illuminates the need for tailored interventions and instructional strategies to cater to the specific needs of different learner subgroups.

Students in profile 1 benefit from a high-quality learning atmosphere, exhibit very low levels of negative emotions and demonstrate a very high willingness to communicate. This profile represents the ideal educational setting, where students are highly engaged and experience minimal emotional barriers to communication. Students in profile 2 have the highest learning atmosphere score but experience moderate negative emotions, which may slightly hinder their willingness to communicate. The presence of moderate negative emotions suggests potential stressors or challenges that, while not debilitating, affect their overall communication willingness. Students in profile 3 experience a slightly lower learning atmosphere and higher negative emotions compared to the other profiles. Despite these challenges, their willingness to communicate remains relatively high. This indicates resilience and the potential effectiveness of support mechanisms that help maintain communication willingness even when the learning atmosphere is not optimal.

The distribution of students across the profiles indicates diverse experiences and emotional states within the educational environment. Understanding these profiles can help educators tailor their approaches to address the specific needs of each group. For Profile 1, maintaining a high learning atmosphere and low negative emotions is crucial. For Profiles 2 and 3, interventions could focus on reducing negative emotions and enhancing the learning atmosphere to boost their willingness to communicate. Strategies such as emotional support programs, stress management workshops, and enhancing classroom environments can help in shifting students from Profiles 2 and 3 towards Profile 1, thereby improving overall educational outcomes.

Furthermore, the exploration of the interrelationships between the learning atmosphere, negative emotions, and willingness to communicate contributes to our understanding of how these factors interact and influence each other. Emotions play a significant role in willingness to communicate fluctuation. This study suggests that negative emotions significantly impede students' WTC. Students experience anxiety, fear of negative evaluation, and frustration in English class, which can markedly hinder the learning process by reducing motivation, confidence, and cognitive functions essential for language learning (Geng and Wang 2022; Wang 2023), thereby obstructing WTC. Thus, Converting negative emotions into positive ones, such as enjoyment, is pivotal in enhancing WTC both at the individual and classroom levels (Khajavy, MacIntyre, and Barabadi 2017). According to Li, Jiang, and Dewaele (2018), FLE can be divided into three aspects, namely, FLE-Private, FLE-Teacher, and FLE-Atmosphere. When students are dissatisfied with their learning environment, they may feel less control over their learning outcomes and perceive less value in communication, potentially further hindering their WTC (Solhi 2024). Conversely, in a positive learning environment, teachers' immediacy and rapport play a critical role in fostering students' WTC (Yang and Lin 2024; Zhi and Wang 2024). Therefore, it is essential for educators to cultivate a positive learning environment through effective communication, supportive feedback, personal connections, and an inclusive, respectful atmosphere (Lee, Shin, and Noh 2023).

The study's findings regarding the role of enjoyment and anxiety on EFL learners' willingness to communicate are particularly significant. Understanding the impact of both positive and negative emotions on communication behaviors can inform the development of strategies to enhance learners' motivation and engagement, thereby fostering a more conducive learning atmosphere. By recognizing the influence of emotions, educators can design activities that promote positive affective states and reduce anxiety, ultimately facilitating learners' willingness to participate actively in EFL learning and communication. Moreover, the examination of how different factors contributing to the learning atmosphere, such as family functioning and adolescent psychiatric symptomatology, affect learners' willingness to communicate highlights the broader contextual influences on FLL. This finding extends beyond the traditional classroom setting and emphasizes the need to consider

multiple environmental factors that shape learners' experiences and psychological well-being. It underscores the significance of a holistic approach to EFL learning, taking into account learners' individual circumstances and their interaction with the larger sociocultural context. The study's identification of distinct student profiles based on learning atmosphere, negative emotions, and willingness to communicate aligns with findings from previous research. Similar to the current findings, previous research has also identified diverse emotional and learning profiles among students. Dowd et al. (2019) highlighted the multidimensional nature of student learning dispositions, emphasizing that understanding these profiles can help tailor educational approaches to meet specific student needs.

Previous studies have also utilized latent profile analysis or cluster analysis to identify distinct profiles or subgroups of foreign language learners based on various factors. For example, studies have identified profiles based on language anxiety, motivation, or learning styles. This consistency suggests that there is substantial heterogeneity among EFL learners, and understanding these profiles is important for catering to their specific needs. Wortha et al. (2019) discussed the role of negative emotions in learning environments, finding that negative emotions can significantly impact learning outcomes. This supports the recommendation to focus on reducing negative emotions in Profiles 2 and 3 to enhance students' willingness to communicate.

Previous research has highlighted the interconnectedness of these variables, although the specific dynamics may vary. For instance, studies have shown that a positive learning atmosphere can reduce anxiety, enhance motivation, and increase willingness to communicate. Additionally, negative emotions, such as anxiety or fear, have consistently been found to hinder willingness to communicate. These findings reinforce the notion that the learning atmosphere can significantly influence learners' emotions and subsequently impact their willingness to communicate. Fazel et al. (2014) noted the effectiveness of social and emotional learning programs in schools, which aligns with the suggested interventions of emotional support programs and stress management workshops to improve the learning atmosphere and reduce negative emotions.

The importance of positive emotions, such as enjoyment, in promoting willingness to communicate aligns with previous research that has emphasized the role of affective factors in EFL learning. Studies have shown that a positive emotional experience can foster a more favorable learning environment, enhance motivation, and increase willingness to communicate. Similarly, the impact of anxiety on communication behaviors is consistent with the literature that highlights the detrimental effects of anxiety on EFL learning and communication. Yang et al. (2022) identified different engagement and burnout profiles among students, indicating that tailored interventions based on specific profiles can effectively address unique student needs, similar to the suggested strategies for shifting students towards Profile 1.

Previous studies have investigated how different aspects of the learning atmosphere, such as teacher-student relationships, classroom climate, and instructional strategies, influence learners' emotional experiences and willingness to engage in communication. This research has consistently found that a supportive and inclusive learning atmosphere positively impacts learners' emotional well-being and their willingness to participate actively in communication activities. Young, Sandman, and Craske (2019) highlighted the importance of emotion regulation strategies in managing both positive and negative emotions. Implementing such strategies can help students in Profiles 2 and 3 better manage their emotions, thus improving their overall learning experience and willingness to communicate. Overall, these comparisons indicate that the findings are consistent with previous research, reinforcing the importance of addressing emotional states and learning environments to improve educational outcomes. While previous studies have explored various factors contributing to the learning atmosphere, such as teacher behaviors or peer interactions, the examination of factors beyond the traditional classroom setting, such as family functioning or psychiatric symptomatology, may provide a broader understanding of contextual influences. Previous research has examined the impact of family support, social relationships, and cultural background on learners' willingness to communicate. These studies emphasize the importance of considering multiple environmental factors that shape learners' experiences.

The new findings from this study provide several novel insights and advancements compared to previous research. Unlike many prior studies that often group students into broader categories, this study provides a detailed breakdown of three distinct student profiles based on specific metrics: learning atmosphere, negative emotions, and willingness to communicate. This granularity allows for more tailored educational interventions. The study highlights a strong correlation between a positive learning atmosphere and a high willingness to communicate, particularly in Profile 1. Previous studies have generally explored these variables separately, but this study emphasizes their interconnectedness and the importance of maintaining a high-quality learning environment to foster communication skills (Romanelli, Bird, and Ryan 2009).

While previous research has acknowledged the impact of negative emotions on learning, this study explicitly identifies the varying levels of negative emotions across different student profiles and their direct impact on willingness to communicate. This nuanced approach underscores the need for targeted emotional support and stress management interventions (Wortha et al. 2019; Young, Sandman, and Craske 2019). The recommendations for tailored interventions, such as emotional support programs and stress management workshops, based on the identified profiles, offer a practical framework for educators. This is a step forward from general recommendations to more precise strategies that can be directly applied in educational settings (Fazel et al. 2014). The concept of shifting students from Profiles 2 and 3 towards Profile 1 by improving the learning atmosphere and reducing negative emotions is a strategic approach that builds on the existing literature but provides a clear pathway for educators to follow. This dynamic view of student profiles is innovative and actionable (Houck et al. 2016). These findings collectively offer a more nuanced understanding of the interplay between the learning atmosphere, negative emotions, and willingness to communicate, providing educators with targeted strategies to improve student outcomes.

Overall, these findings contribute to our understanding of the intricate relationships between learning atmosphere, negative emotions, and willingness to communicate among EFL learners. They underscore the importance of creating positive and supportive learning environments that address learners' emotional needs, enhance their motivation, and promote their willingness to engage in communication. By recognizing the heterogeneity of learners' experiences and uncovering the underlying dynamics, these findings can inform the development of targeted interventions, instructional strategies, and policy recommendations aimed at optimizing EFL learning environments and facilitating learners' learning journey.

Conclusion

In conclusion, the findings of the study investigating latent profiles in the dynamics of learning atmosphere, negative emotions, and willingness to communicate among EFL learners offer a rich understanding of the intricate interplay between these factors in language learning environments. By identifying distinct latent profiles or subgroups within the EFL learner population based on their experiences, the study highlights the diverse ways in which learners interact with and respond to the learning atmosphere, negative emotions, and willingness to communicate.

The exploration of the interrelationships between the learning atmosphere, negative emotions, and willingness to communicate sheds light on the complex dynamics that shape learners' language learning experiences. By uncovering how these factors interact and influence each other, the study provides a deeper understanding of the mechanisms underlying learners' emotional responses and their communication behaviors. These insights can inform educators and policymakers in designing targeted interventions and support systems to enhance EFL learners' learning experiences (Gao, Wang, and Fan, 2024; Yang and Lin 2024). Moreover, the investigation into the role of enjoyment and anxiety on EFL learners' willingness to communicate underscores the significance of affective factors in shaping learners' engagement and motivation. By emphasizing the impact of both positive and negative emotions on communication behaviors, the study underscores the importance of creating supportive and inclusive learning environments that promote positive emotional

experiences and reduce barriers to effective communication. The study's examination of how the learning atmosphere influences learners' emotional experiences and willingness to engage in communication activities highlights the pivotal role of the classroom environment and teaching methodologies in shaping EFL learners' learning journey. Understanding the impact of these factors can guide practitioners in creating stimulating and nurturing learning environments that optimize learners' emotional well-being and foster their communication skills.

Lastly, the study's exploration of how factors beyond the classroom setting, such as family functioning and adolescent psychiatric symptomatology, influence EFL learners' willingness to communicate broadens our understanding of the contextual influences on language learning. By considering these external factors, educators can adopt a more holistic approach to supporting learners and addressing their individual needs within and outside the classroom. Overall, these findings provide valuable insights into the multifaceted nature of EFL learning environments and their profound impact on learners' emotional experiences and communication behaviors. By recognizing the complexity of these dynamics, stakeholders in language education can develop informed strategies and interventions to create more enriching, supportive, and effective leaning environments for diverse populations of EFL learners.

Implications

The findings of the study investigating latent profiles in the dynamics of learning atmosphere, negative emotions, and willingness to communicate among EFL learners hold both theoretical and practical implications that can significantly influence language education practices and policy decisions. Here are some key theoretical and practical implications based on the findings:

Theoretical implications

The identification of distinct latent profiles among EFL learners contributes to the advancement of theoretical models in language learning and psychology. This enhances our understanding of the heterogeneous nature of learners' experiences and behaviors within different EFL learning contexts. Understanding the interrelationships between the learning atmosphere, negative emotions, and willingness to communicate elucidates the complex interactions among these variables. This deeper comprehension can enrich existing theories on EFL learning motivation, affective factors, and communication behaviors. The investigation into the role of enjoyment and anxiety in EFL learners' willingness to communicate highlights the significance of affective factors in learning processes. This emphasizes the need to integrate emotional considerations into theoretical frameworks related to EFL learning and communication.

Practical implications

The identification of latent profiles enables educators to tailor interventions and support strategies based on the specific needs and characteristics of different learner subgroups. This personalized approach can enhance learning outcomes and foster a more inclusive and supportive learning environment. Understanding how the learning atmosphere influences learners' emotional experiences and willingness to communicate underscores the importance of creating positive and engaging learning environments. Educators can use these insights to design classroom practices, teaching methodologies, and interactions that promote a supportive atmosphere conducive to EFL learning. Insights into how family functioning and other external factors impact learners' willingness to communicate empower educators to consider a holistic approach to language education. By recognizing the broader contextual influences on learners, educators can collaborate with families, mental health professionals, and other stakeholders to provide comprehensive support. These findings can inform professional development programs for language educators, helping them

develop skills and strategies to address diverse emotional needs and communication barriers among learners. Educators can learn to create inclusive and motivating learning environments that foster communication skills and emotional well-being. Policymakers can utilize these findings to inform decisions related to curriculum design, teacher training, and resource allocation to enhance learning experiences for diverse learner populations. Policies that prioritize positive learning atmospheres and emotional support can lead to more effective language education programs. The theoretical and practical implications of the study's findings underscore the importance of considering the complex dynamics of the learning atmosphere, negative emotions, and willingness to communicate in language education. By integrating these insights into educational practices and policies, stakeholders can create more supportive, inclusive, and effective EFL learning environments that nurture learners' emotional well-being and communication skills.

Limitations

The study's sample size might be limited, potentially affecting the generalizability of findings to broader EFL learner populations. The reliance on self-reported measures for assessing variables like negative emotions and willingness to communicate may introduce biases and inaccuracies due to social desirability or recall issues. If the study adopts a cross-sectional design, it may only capture a snapshot of participants' experiences, limiting the ability to draw causal conclusions or understand changes over time. There could be limitations in the validity of the measurement tools used to assess latent profiles, learning atmosphere, negative emotions, and willingness to communicate, impacting the accuracy of the results. The study's findings may not fully account for cultural differences in the perception and expression of emotions, as well as in the interpretation of the learning atmosphere and willingness to communicate. Certain variables not accounted for in the study, such as prior EFL learning experiences or individual differences in coping mechanisms, could influence the observed relationships among the studied variables.

Suggestions for future studies

Conduct longitudinal studies to track changes in latent profiles, learning atmosphere, negative emotions, and willingness to communicate over time. This would provide insights into the temporal dynamics and potential causal relationships among these variables. Employ a mixed-methods approach combining quantitative latent profile analysis with qualitative data collection methods such as interviews or focus groups. This would offer a comprehensive understanding of the complex interplay between latent profiles and qualitative experiences of EFL learners. Conduct cross-cultural comparative studies to explore how cultural factors influence the dynamics of learning atmosphere, negative emotions, and willingness to communicate among EFL learners. Comparing different cultural contexts could reveal unique patterns and shed light on culturally specific interventions. Implement intervention studies aimed at modifying the learning atmosphere to promote positive emotions and enhance willingness to communicate among EFL learners. This could involve implementing strategies to reduce negative emotions and enhance motivation within the learning environment. Investigate the role of teacher training programs in fostering a conducive learning atmosphere that supports positive emotions and facilitates effective communication among language learners. This could involve examining the impact of teacher behaviors and classroom management strategies on student experiences.

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